

國立清華大學 電機工程學系

114 學年度第二學期

EE-6250 超大型積體電路測試 VLSI Testing

Homework #3 (佔學期總成績 10 分)

Due Date : **23:59pm, June 15, 2026**, (逾時不收)

Late Homeworks will NOT be accepted !

Submit to <https://eeclass.nthu.edu.tw/> 作業區

1. (10%) Consider the Built-In Speed Grading for a logic circuit.
 - (a) Derive the logic circuits of an unsigned 6-bit adder. (You can use Design Compiler if you know how to use it.
 - (b) **Insert the BIST circuit into your adder** by using a 4-stage **Linear Feedback Shift-Register (LFSR)** as the automatic pattern generator of 15 pseudo-random patterns, and a **Multiple-Input Shift-Register (MISR)** as the signature analyzer. Draw the architecture of your BISTed design, and report its golden signature when the circuit is fault-free.
 - (c) Set up the simulation environment to verify that your adder operates at 100 MHz. Report the simulation waveforms of important input/output signals, and the signature produced at this speed.
 - (d) Find a procedure to gauge the **maximum operating speed** of your BISTed adder design by running **simulated Built-In Self-Test** multiple times. During the procedure, use 100ps as the time resolution when you tune the clock cycle time for the simulation. (Note: The procedure could be either manual or automated by a script. You are advised to write a brief paragraph describing how you solved this problem.
 - (e) Replace your circuit under test with an unsigned **12-bit adder**. Report its maximum operating speed again by repeating the procedure in (d).

Note: 繳交資料: Combine all of the following items into a single PDF file for submission to our 【清華大學-數位學習平台】 (<https://eeclass.nthu.edu.tw/>). On top of the combined PDF file, there should be a cover page with your 系所，中英文姓名，學號等資訊.

- In addition to the answers to the above questions, you should include your (1) Verilog Code, (2) testbench, or (3) simulation waves produced in (a)-(d). For problem (e), you can report just the final result, skipping the intermediate results.